arXiv:2309.08532v3 [cs.CL] 1 May 2025

EVOPROMPT: CONNECTING LLMS WITH EVOLUTIONARY ALGORITHMS YIELDS POWERFUL PROMPT OPTIMIZERS

Qingyan Guo[12†*], **Rui Wang**[2†], **Junliang Guo**[2], **Bei Li**[23], **Kaitao Song**[2], **Xu Tan**[2‡], **Guoqing Liu**[2], **Jiang Bian**[2], **Yujiu Yang**[1‡]
[1]Tsinghua University [2]Microsoft Research [3]Northeastern University
gqy22@mails.tsinghua.edu.cn, libei_neu@outlook.com,
{ruiwa,junliangguo,kaitaosong,xuta,guoqingliu,jiabia}@microsoft.com
yang.yujiu@sz.tsinghua.edu.cn

ABSTRACT

Large Language Models (LLMs) excel in various tasks, but they rely on carefully crafted prompts that often demand substantial human effort. To automate this process, in this paper, we propose a novel framework for discrete prompt optimization, called EVOPROMPT, which borrows the idea of evolutionary algorithms (EAs) as they exhibit good performance and fast convergence. To enable EAs to work on discrete prompts, which are natural language expressions that need to be coherent and human-readable, we connect LLMs with EAs. This approach allows us to simultaneously leverage the powerful language processing capabilities of LLMs and the efficient optimization performance of EAs. Specifically, abstaining from any gradients or parameters, EVOPROMPT starts from a population of prompts and iteratively generates new prompts with LLMs based on the evolutionary operators, improving the population based on the development set. We optimize prompts for both closed- and open-source LLMs including GPT-3.5 and Alpaca, on 31 datasets covering language understanding, generation tasks, as well as BIG-Bench Hard (BBH) tasks. EVOPROMPT significantly outperforms human-engineered prompts and existing methods for automatic prompt generation (e.g., up to 25% on BBH). Furthermore, EVOPROMPT demonstrates that connecting LLMs with EAs creates synergies, which could inspire further research on the combination of LLMs and conventional algorithms. Our code is available at https://github.com/beeevita/EvoPrompt.

1 INTRODUCTION

Large language models (LLMs) show remarkable performance on multiple natural language processing (NLP) tasks (Touvron et al., 2023; Ouyang et al., 2022). To adapt to downstream tasks, simply adding an instruction to the input text, also called discrete prompt, steers LLMs to carry out the desired task with negligible impact on computational cost (Liu et al., 2023). Such approach also eliminates the need for all the parameters and gradients in LLMs, making it suitable for LLMs with block-box APIs such as GPT-3 and GPT-4 (Brown et al., 2020; OpenAI, 2023). Despite the convenience, the performance of the LLMs towards a certain task is significantly influenced by the prompt (Liu et al., 2023; Zhu et al., 2023). Accordingly, the key challenge of this approach lies in the design of the prompt, which has emerged as a crucial technique known as prompt engineering (Zhou et al., 2022). Given the wide variation in prompts across language models and tasks, the prompt design typically requires substantial human effort and expertise with subjective and relatively limited guidelines (Mishra et al., 2022a;b; Liu et al., 2023; Zamfirescu-Pereira et al., 2023; Wang et al., 2023).

[*]Work done during an internship at Microsoft Research Asia.
[†]Equal Contribution.
[‡]Corresponding Author.

To alleviate human effort on discrete prompt design, previous approaches usually rely on access to the token probabilities from the output layer of LLMs, which may not always be accessible through APIs (Deng et al., 2022; Zhang et al., 2023a). Some recent works consider enumerating diverse prompts and selecting the best ones (Zhou et al., 2022; Jiang et al., 2020), or modifying current prompts to improve them (Guo et al., 2023; Prasad et al., 2022; Pryzant et al., 2023). Such approaches either emphasize *exploring* diverse prompts, which may lead to indecisiveness and wasted resources, or focus on *exploiting* upon the current identified good prompts, which may result in stagnation and confine the search to local optima. Several conventional derivative-free algorithms are well-designed and strike a good balance between *exploration* and *exploitation* (Conn et al., 2009; Rios & Sahinidis, 2013). Among these, evolutionary algorithms (EAs) stand out as they are simple and efficient, as well as suitable for discrete prompt optimization (Storn & Price, 1997; Brest et al., 2006; Zhang & Sanderson, 2009; Vesterstrom & Thomsen, 2004). Sequences of phrases in prompts can be regarded as gene sequences in typical EAs, making them compatible with the natural evolutionary process.

In this paper, we borrow the idea of EAs and propose a discrete prompt tuning framework, EVOPROMPT. While evolutionary operators in EAs are typically designed for sequences, they tend to independently alter tokens to generate new candidate solutions. Unfortunately, this approach ignores the connections among tokens, which is crucial for maintaining coherence and readability in prompts. Taking advantage of LLMs' expertise in NLP and the exceptional optimization capabilities of EAs, we connect these two approaches, where LLMs generate new candidate prompts following evolutionary operators, and EAs guide the optimization process to retain the optimal prompts.

Specifically, based on several initial prompts, we utilize LLMs to act as evolutionary operators to generate new prompt candidates, and the prompt with better performance on the development set is preserved. The above operations upon the updating population are iteratively applied to improve the quality. By elaborately designing the evolutionary operators and adjusting the update strategy, EVOPROMPT can be instantiated with various types of EAs. We optimize the prompts for two different LLMs (i.e., Alpaca (Taori et al., 2023), and GPT-3.5 (Brown et al., 2020)) on a diverse range of neural language understanding and generation tasks, as well as challenging BIG-Bench tasks, using a total of 31 datasets. EVOPROMPT consistently gets better prompts compared with both manually designed ones and previous automatic prompt generation methods. The main contributions of this paper include:

- We propose a novel framework for automatic discrete prompt optimization connecting LLMs and EAs, called EVOPROMPT, which enjoys the following advantages: **1)** It does not require access to any parameters or gradients of LLMs; **2)** It strikes a balance between *exploration* and *exploitation* leading to better results; **3)** The generated prompts are human-readable.
- Experiments conducted on 31 datasets demonstrate the effectiveness of EVOPROMPT compared with crafted prompts, as well as existing methods. We release the optimal prompts obtained by EVOPROMPT for these common tasks such as sentiment classification, topic classification, subjectivity classification, simplification, summarization and reasoning.
- We demonstrate that LLMs are capable of implementing multiple types of EAs provided with appropriate instructions. We hope that our explorations will inspire further investigations on the combination of LLMs and conventional algorithms, paving the way for new and innovative applications of LLMs.

2 RELATED WORKS

Prompts in LLMs Prompting is an efficient method for employing LLMs in specialized tasks. However, the performance is heavily influenced by the choice of the prompt. Recently, automatic prompt optimization has obtained wide attention. Continuous prompt-based methods, which only tune parameters of some input tokens (Li & Liang, 2021; Liu et al., 2021b;a; Zhang et al., 2021) attract lots of attention. In spite of their effective performance, two drawbacks of such paradigms can not be ignored: **1)** The optimization of continuous prompts requires parameters of LLMs that are inaccessible for black-box APIs. **2)** Soft prompts often fall short of interpretability (Lester et al., 2021). Discrete prompts, simply adding several discrete tokens, such as "It was" (Schick & Schütze, 2021), or task-specific descriptive instructions, such as "Classify the comment into positive or negative.", to the input text, can offer an interactive interface to humans with better interpretability and show promising performance in various NLP tasks (Liu et al., 2023).

Discrete Prompts Various approaches have been proposed for automatic discrete prompt searching and generation (Shin et al., 2020; Shi et al., 2022; Wallace et al., 2019; Deng et al., 2022; Zhang et al., 2023a), while these methods still rely on the gradients or the token probabilities from the output layer. More recently, considering the high variance of different prompts for downstream tasks, some works focus on *exploration* by enumerating and selecting the best prompt from a number of candidates, mainly augmented by re-sampling (Zhou et al., 2022; Jiang et al., 2020). Approaches based on prompt edit (Zhang et al., 2023a; Prasad et al., 2022) emphasize *exploitation*, which may potentially lead to local optima. Another approach collects the incorrectly predicted cases and analyzes the corresponding root cause to improve existing prompts (Pryzant et al., 2023; Guo et al., 2023), which also emphasizes *exploitation*. Additionally, such approaches are constrained to tasks with standard answers and cannot be directly applied to generation tasks. Our proposed EVOPROMPT empowered with evolutionary algorithms strikes a balance between *exploration* and *exploitation* without requiring any parameters or gradients.

LLMs and Optimization Algorithms LLMs demonstrate the potential to serve as black-box optimizers (Zheng et al., 2023); however, this black-box approach lacks explainability. Some works have revealed that LLMs have the capability to imitate specific operations in conventional algorithms. For instance, LLMs can perform "Gradient Descent" in discrete space by collecting incorrectly predicted samples (Pryzant et al., 2023; Guo et al., 2023). Meanwhile, it has been demonstrated that LLMs can imitate the mutation (Lehman et al., 2022) or crossover (Meyerson et al., 2023) operator in the genetic algorithm (GA). Chen et al. (2023) further integrates LLMs and GA for neural architecture search, while Lanzi & Loiacono (2023) introduce a similar approach to game design. Our work has taken a significant step forward by proposing a general framework that connects LLMs with evolutionary algorithms, which can be instantiated to a diverse range of evolutionary algorithms through customization of evolutionary and selection processes, thereby broadening its applicability and potential influence in the domain. We aspire this work to inspire broader applications of combining LLMs and conventional algorithms.

3 AUTOMATIC DISCRETE PROMPT OPTIMIZATION

Algorithm 1 Discrete prompt optimization: EVOPROMPT

Require: Initial prompts $P_0 = \{p_1, p_2, \ldots, p_N\}$, size of population N, a dev set $\mathcal{D}$, $f_{\mathcal{D}}(\cdot)$ denotes the score of a prompt on the desired LLM evaluated on $\mathcal{D}$, a pre-defined number of iterations T, carefully designed evolutionary operators to generate a new prompt Evo$(\cdot)$

1: **Initial evaluation scores**: $S_0 \leftarrow \{s_i = f_{\mathcal{D}}(p_i) | i \in [1, N]\}$
2: **for** $t = 1$ to T **do**
3: **Selection**: select a certain number of prompts from current population as parent prompts $p_{r_1}, \ldots, p_{r_k} \sim P_{t-1}$
4: **Evolution**: generate a new prompt based on the selected parent prompts by leveraging LLM to perform evolutionary operators $p'_i \leftarrow$ Evo$(p_{r_1}, \ldots, p_{r_k})$
5: **Evaluation**: $s'_i \leftarrow f(p'_i, \mathcal{D})$
6: **Update**: $P_t \leftarrow \{P_{t-1}, p'_i\}$ and $S_t \leftarrow \{S_{t-1}, s'_i\}$ based on the evaluation scores
7: **end for**
8: **Return** the best prompt, p^*, among the final population P_T: $p^* \leftarrow argmax_{p \in P_T} f(p, \mathcal{D})$

Current advanced LLMs are typically interacted via black-box APIs, while the gradients and parameters are inaccessible. Evolutionary algorithms (EAs) are derivative-free algorithms with exceptional accuracy and rapid convergence. Accordingly, we consider introducing EAs into discrete prompt optimization. However, to generate new candidate solutions, evolutionary operators typically edit the elements in current solutions independently, without considering the connections between them. This makes it challenging to apply evolutionary operators on discrete prompts, which require coherence and readability. To address this challenge, we propose a synergistic approach that connects the natural language processing expertise of LLMs with the optimization capabilities of EAs, called EVOPROMPT. Specifically, LLMs generate new candidate prompts based on evolutionary operators, while EAs guide the optimization process to find the optimal prompts.